

User-Mode Linux



This article gives you hands-on experience in setting up a User-Mode Linux (UML) kernel and getting it up on a running Linux OS. It shows how to share files between the host Linux and guest Linux, via the network and other methods. It covers building a custom kernel, building modules for the UML kernel, inserting them into the running UML kernel, and debugging the kernel and modules with GDB. This article assumes that the reader is an advanced Linux user, programmer or administrator, with hands-on knowledge of Linux, and a basic understanding of kernel compilation.

UML gives you the advantage of running Linux over a Linux distribution, without the need of privileged access. It is run in the form of an unprivileged user program, giving the end user the power to play with the OS. The Linux kernel, once compiled to the UML architecture, creates a machine-dependent binary which can execute itself, and launch the UML kernel.

UML was developed by Jeff Dike, and has been part of the vanilla Linux kernel since version 2.6.0. It is very useful for kernel developers to quickly test new code; and for administrators, to build sandbox Linux virtual machines and honey-pots, while deploying new services without disturbing the production

environment. Most steps mentioned in this document are distribution-agnostic, and can be tried on any Linux machine with the x86 or x86_64 architectures.

Figure 1 depicts a conceptual layout of UML in relation to the hardware, host kernel and user-space.

Requirements for setting up UML

The basic requirements for setting up UML are:

- Access to a Linux machine with x86 or x86_64 architecture (with or without root access)
- The Linux kernel build environment pre-installed (GCC, *make*, etc)
- A downloaded kernel source tarball (version 2.6.x) from kernel.org (2.6.35-rc3 is used in this article)